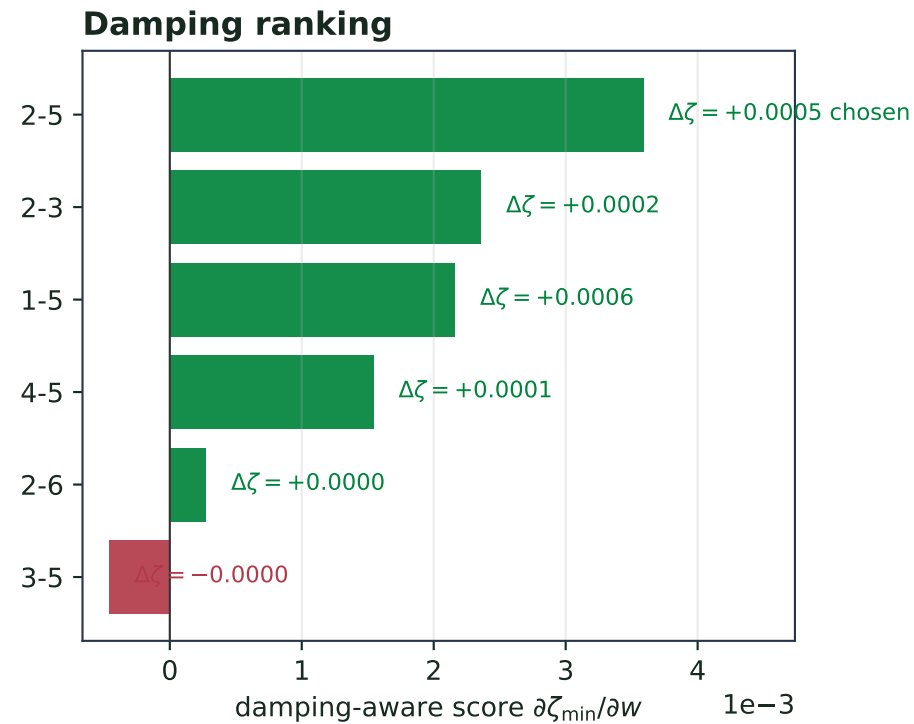
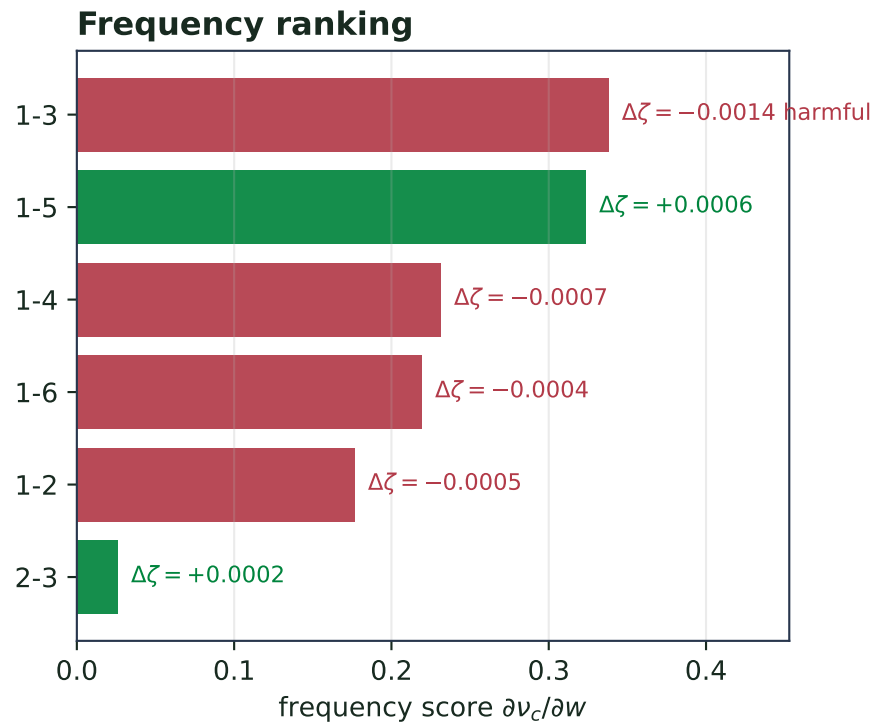


A real reversal: the cheapest frequency heuristic picks the harmful line



Seed 727986944, moderate non-proportional damping. Line 1-3 raises critical modal stiffness by 0.0931 but lowers full-QEP damping margin by -0.0014 under a 10% reinforcement.